

STEP 1 Confirm the print scale

Print this page first. Measure the ruler below with a steel rule or calipers.

It must read exactly 100.0 mm. If it reads 97 mm or 104 mm, the driver rescaled the page:

set scaling to None / 100% / Actual size, turn off Fit to page and Shrink oversized pages, then reprint.

Do not print the board pages until this ruler measures correctly.



STEP 2 Print the boards

Matte paper or matte adhesive vinyl only. Gloss produces specular highlights that destroy sub-pixel corner refinement under the workshop lighting the rig will actually face.

Laser preferred over inkjet: crisper black-to-white transitions at the square boundary.

Print in true black, greyscale off, toner saver off, no watermark or header from the driver.

STEP 3 Mount before measuring

Bond each board to a rigid flat substrate: dibond, 2 to 3 mm aluminium, or 3 mm acrylic.

Foamboard is acceptable short term, cardboard is not. Paper curl is a systematic pose error,

not noise: it biases every observation in the same direction and will not average out.

Apply adhesive over the full area, not at the corners, and roll out air bubbles.

STEP 4 Measure the mounted board and record the result

Measure with calipers ACROSS FOUR SQUARES and divide by four. This divides your reading error by four and is the single cheapest accuracy gain available on this rig.

Write the measured value into rig.yaml. Never use the nominal value from the generator:

printers scale by a few tenths of a percent, and scale error propagates linearly into translation,

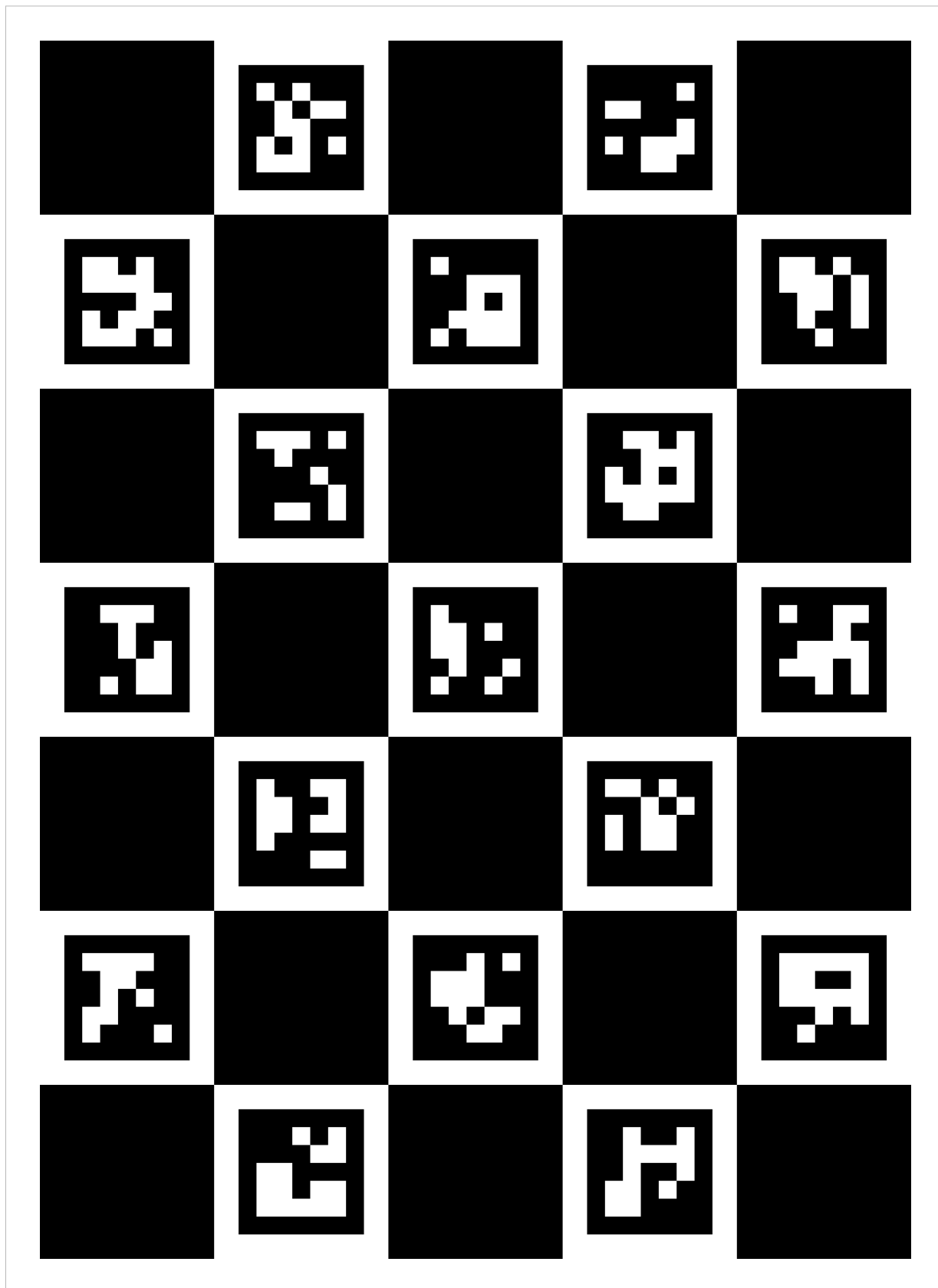
so a 0.3% square error becomes a 0.3% range error and lands directly on the component position.

Record the measurement uncertainty too. It is the input to the Monte Carlo uncertainty region.

ID ALLOCATION (disjoint ranges, no board can be confused with another)

Board	Layout	IDs	Square	Extent
calibration_board	5 x 7	0-16	30.0 mm	150 x 210 mm
rig_board_a	4 x 6	20-31	25.0 mm	100 x 150 mm
rig_board_b	4 x 6	35-46	25.0 mm	100 x 150 mm

Spare IDs 17-19, 32-34 and 47-99 are unallocated. Keep them free for later rig extensions.

**NOMINAL SPECIFICATION (not a measurement)**

Dictionary: DICT_5X5_100

Layout: ChArUco 5 x 7 squares

Marker IDs: 0 to 16 (17 markers)

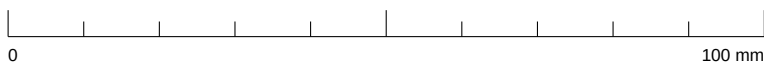
rig.yaml key: calibration_board

Square side: 30.00 mm

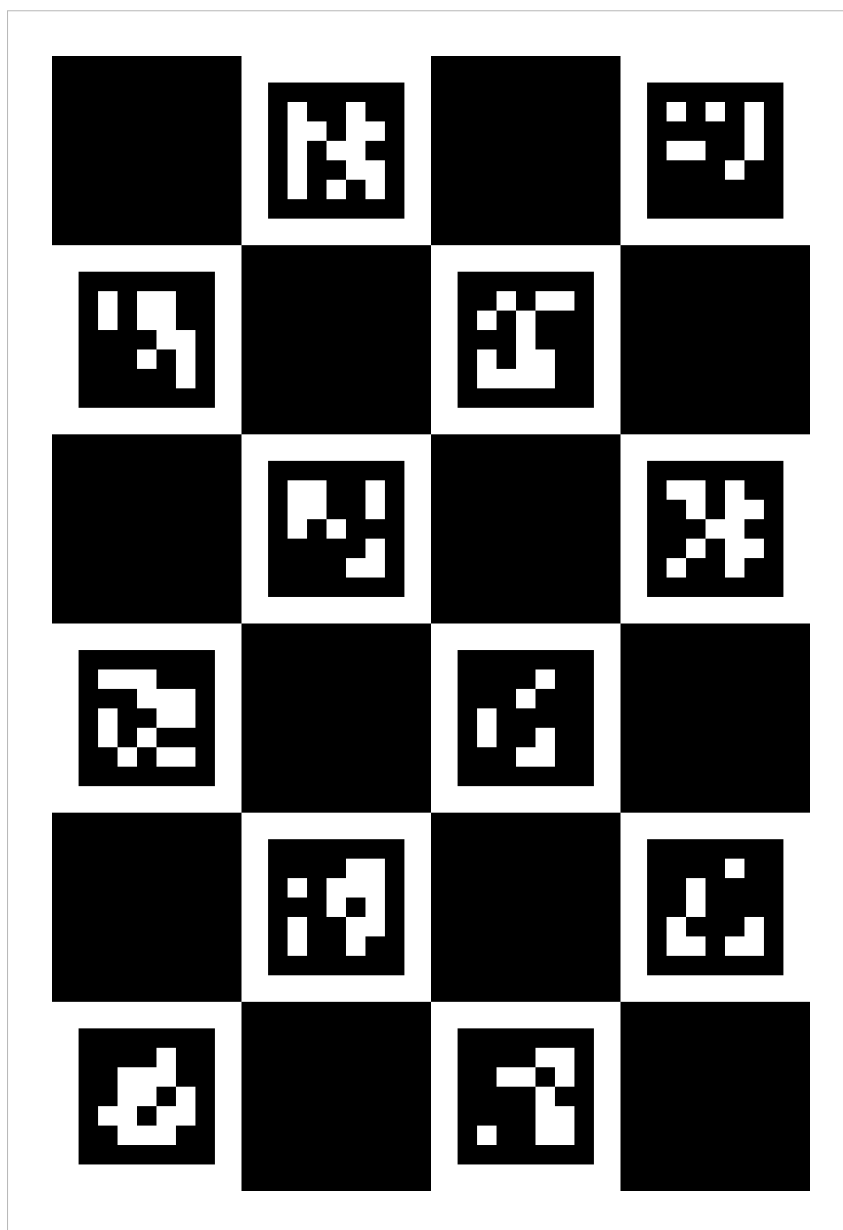
Marker side: 21.60 mm

Board extent: 150.0 x 210.0 mm

Chessboard corners: 24



Measured square side: mm Uncertainty: +/- mm Date:

**NOMINAL SPECIFICATION (not a measurement)**

Dictionary: DICT_5X5_100

Layout: ChArUco 4 x 6 squares

Marker IDs: 20 to 31 (12 markers)

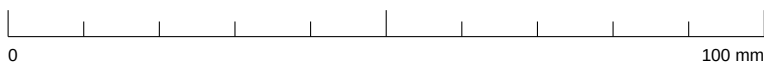
rig.yaml key: rig_board_a

Square side: 25.00 mm

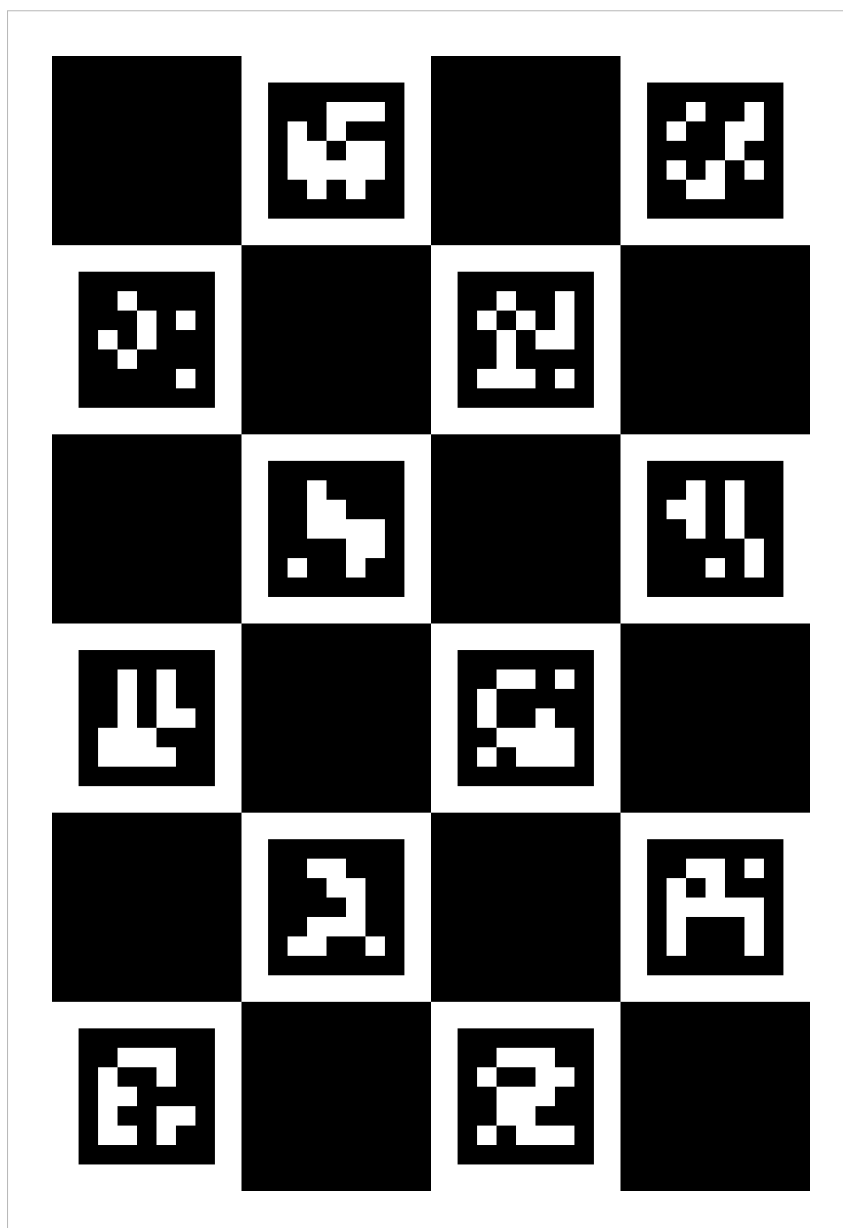
Marker side: 18.00 mm

Board extent: 100.0 x 150.0 mm

Chessboard corners: 15



Measured square side: mm Uncertainty: +/- mm Date:

**NOMINAL SPECIFICATION (not a measurement)**

Dictionary: DICT_5X5_100

Layout: ChArUco 4 x 6 squares

Marker IDs: 35 to 46 (12 markers)

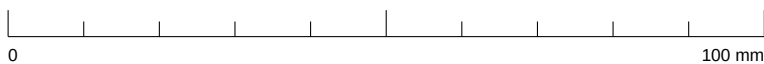
rig.yaml key: rig_board_b

Square side: 25.00 mm

Marker side: 18.00 mm

Board extent: 100.0 x 150.0 mm

Chessboard corners: 15



Measured square side: mm Uncertainty: +/- mm Date: